

Soham Jana

Assistant Professor, University of Notre Dame

Department of Applied and Computational Mathematics and Statistics

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Research Interests

Theoretical and methodological aspects of high-dimensional statistics, neural networks, mixture modeling, distance-based estimators.

Education

Ph.D. in Statistics and Data Science, Yale University 2022

Advisor: Yihong Wu

Thesis: Learning non-parametric and high-dimensional distributions via information-theoretic methods

Master of Statistics (First class with distinction), Indian Statistical Institute 2017

Specialization: Theoretical Statistics

Dissertation: Characterization of single-integral non-kernel divergences

Advisor: Ayanendranath Basu

Bachelor of Statistics (First class with distinction), Indian Statistical Institute 2015

Employment

University of Notre Dame 2024–Present

Assistant Professor, Department of Applied and Computational Mathematics and Statistics

Princeton University 2022–2024

Postdoctoral Researcher, Department of Operations Research and Financial Engineering

Hosts: Sanjeev Kulkarni, Jianqing Fan

Researcher, First Republic Bank Research and Lifelong Learning Program 2022–2023

Lecturer Spring 2023, Fall 2023

Grants and Awards

Kaneb Center Course Design Academy, University of Notre Dame 2024–2025

Award Amount: \$5000

Publications

* Authors not in alphabetical order † Equal contribution

Journal Publications

1. **Jana, S.**, Yang, K., & Kulkarni, S. (2026).*
Adversarially robust clustering with optimality guarantees.
IEEE Transactions on Information Theory, 72(1), 478–500.
2. **Jana, S.**, Polyanskiy, Y., & Wu, Y. (2025).
Optimal empirical Bayes estimation for the Poisson model via minimum-distance methods.
Information and Inference, Volume 14, Issue 4.
3. **Jana, S.**, Fan, J., & Kulkarni, S. (2025).*
A provable initialization and robust clustering method for general mixture models.
IEEE Transactions on Information Theory, vol. 71, no. 9.
4. **Jana, S.**, Li, H., Yamada, Y., & Lindenbaum, O. (2023).
Support recovery with stochastic gates: theory and application for linear models.
Signal Processing, 213.
5. Han, Y., **Jana, S.**, & Wu, Y. (2023).
Optimal prediction of Markov chains with and without spectral gap.
IEEE Transactions on Information Theory, 69(6).
6. **Jana, S.**, & Basu, A. (2019).*
A characterization of all single-integral, non-kernel divergence estimators.
IEEE Transactions on Information Theory, 65(12).

Conference Publications

1. **Jana, S.**, Polyanskiy, Y., Teh, A., & Wu, Y. (2023).
Empirical Bayes via ERM and Rademacher complexities: the Poisson model.
Conference on Learning Theory (COLT), PMLR 195:5199–5235.
2. Han, Y., **Jana, S.**, & Wu, Y. (2021).
Optimal prediction of Markov chains with and without spectral gap.
Neural Information Processing Systems (NeurIPS).
3. **Jana, S.**, Polyanskiy, Y., & Wu, Y. (2020).
Extrapolating the profile of a finite population.
Conference on Learning Theory (COLT).

Preprints

1. Xing, H.[†], **Jana, S.**[†], & Maleki, A. (2025).*
Minimax analysis of estimation problems in coherent imaging.
arXiv:2508.18503.
2. Fan, J., **Jana, S.**, Kulkarni, S., & Yin, Q. (2025).
Factor informed double deep learning for average treatment effect estimation.
arXiv:2508.17136.
3. Chen, X.[†], **Jana, S.**[†], Metzler, C., Maleki, A., & Jalali, S. (2025).*
Multilook coherent imaging: theoretical guarantees and algorithms.
arXiv:2505.23594.
4. Tang, S., **Jana, S.**, & Fan, J. (2024).
Factor adjusted spectral clustering for mixture models.
Under major revision at *Journal of the American Statistical Association*.

Conferences, Workshops, and Invited Talks

- Statistical Learning and Data Science: Inference and Intelligence, New York, USA (November 2026)
- World Meeting of the International Society for Bayesian Analysis, Nagoya, Japan (June 2026)
- IMSI Workshop on Reinforcement Learning from Offline Data and Human Feedback, Chicago, USA (April 2026)
- Joint Statistical Meetings, Nashville, TN, USA (August 2025)
- IMS New Researchers Conference, Nashville, TN, USA (July–August 2025)
- International Webinar on Recent Trends in Statistical Theory and Applications, Kerala, India (July 2025)
- International Indian Statistical Association, Lincoln, NE, USA (June 2025)
- International Indian Statistical Association, Cochin, Kerala, India (December 2024)
- Joint Statistical Meetings, Portland, OR, USA (August 2024)
- University of Notre Dame Statistics Department Seminar, Notre Dame, IN, USA (February 2024)
- University of Wisconsin–Madison Statistics Department Seminar, Madison, WI, USA (February 2024)
- University of Texas at Dallas Statistics Department Seminar, Richardson, TX, USA (January 2024)
- Indian Statistical Institute ISRU Department Seminar, Kolkata, India (July 2023)
- Conference on Learning Theory (COLT), Bangalore, India (July 2023)
- Neural Information Processing Systems (NeurIPS), Virtual (December 2021)
- Conference on Learning Theory (COLT), Graz, Austria (July 2020)

Teaching

University of Notre Dame

- ACMS 30530 – Introduction to Probability (Fall 2024, 2025, Spring 2026)
- ACMS 70100 – Machine Learning Methods for High-dimensional Data (Spring 2026)
- ACMS 80870 – Modern Machine Learning Techniques with Application (Spring 2025)

Princeton University

- ORF 309/ENG 309/MAT 380 – Probability and Stochastic Systems (Spring 2023)
- ORF 570 – Statistical Machine Learning (Fall 2023)

Professional Service

Paper reviews (29): Annals of Statistics (3), Journal of the American Statistical Association (2), IEEE Transactions on Information Theory (5), Journal of the Royal Statistical Society (2), IEEE Wireless Communications Letters (1), IEEE ISIT (1), Electronic Journal of Statistics (2), Stat (1), Algorithmic Learning Theory (7), Bernoulli (1), Statistica Sinica (1), Journal of Statistical Planning and Inference (1), Biometrics (1), SIAM Journal on Mathematics of Data Science (1).

Conference Organization

- CFE-CMStatistics Conference (December 2025): Session organizer – Recent advances in Causal Inference
- Joint Statistical Meetings (August 2025): Session chair – New Advances in Optimization Algorithms for Causal Discovery
- Joint Statistical Meetings (August 2024): Session chair – New Advances in Nonparametric Hypothesis Testing - Part I
- Joint Statistical Meetings (August 2024): Session chair – New Developments in Non-Euclidean Statistics
- IEEE Conference on Information Sciences and Systems (March 2024): Session chair – Machine learning and statistical inference

Community Service

- Teaching at Math Circle, Notre Dame (Spring 2025, Fall 2025): Promoting STEM education among school children
- Yale S&DS M.A. admission committee (2021): handled 150+ applications
- Yale S&DS graduate reading group (2020): co-organizer
- Yale Women in Data Science (WiDS) workshop (2020): mentor
- South Asian Graduate and Professional Association (SAGA), Yale (2018–2021): Treasurer and organizer

Other Awards

INSPIRE Scholarship, Government of India (2012–2017)
INMO Merit Certificate (Top 75, 2012)